

1. (10 points) Suppose that the vector $(x, 20, 26)$ is in the span of $(1, -1, 3)$ and $(2, -1, 4)$. Find x .

2. Let A be a 4×5 matrix, and let $\mathbf{b}_0 \in \mathbb{R}^4$. Suppose that $A\mathbf{x} = \mathbf{b}_0$ has standard solution

$$\mathbf{x} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -2 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} -3 \\ 0 \\ 0 \\ -4 \\ 1 \end{bmatrix}.$$

(Note that there is no variable in front of the first vector.)

- (a) (3 points) Fill in the blank (no explanation needed): $\text{rank}(A) = \underline{\hspace{2cm}}$.
- (b) (2 points) Circle one (no explanation needed): Every vector $\mathbf{b} \in \mathbb{R}^4$ is a linear combination of $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3, \mathbf{a}_4, \mathbf{a}_5$, where $\mathbf{a}_j$ denotes the j -th column of A .

True

False

- (c) (5 points) Find scalars $c_1, c_2, c_3, c_4 \in \mathbb{R}$ such that $\mathbf{a}_5 = c_1\mathbf{a}_1 + c_2\mathbf{a}_2 + c_3\mathbf{a}_3 + c_4\mathbf{a}_4$.